

NE 523: Final Exam

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1 Adapted Problem from Lewis and Miller

Before we begin, I will use the subscript p to denote a quantity associated with the isotropic point source and the subscript d to denote a quantity associated with the distributed source.

Additionally, I will take the inner product to be the following as we are working with a mono-energetic source.

$$\langle \cdot \rangle = \int_{4\pi} d\hat{\Omega} \int_V dV \cdot \quad (1)$$

Equivalence of q_p and $q_p^\dagger$

1. To begin, let's start by investigating the forward equation for the point source.

$$\hat{\Omega} \cdot \nabla \psi_p(\mathbf{r}, \hat{\Omega}) + \sigma(\mathbf{r})\psi_p(\mathbf{r}, \hat{\Omega}) = \int_{4\pi} d\hat{\Omega}' \sigma_s(\mathbf{r}, \hat{\Omega}' \cdot \hat{\Omega}) \psi_p(\mathbf{r}, \hat{\Omega}') + \frac{S_0}{4\pi} \delta(\mathbf{r} - \mathbf{r}_0) \quad (2)$$

2. Because the source and scattering are isotropic, we know the angular flux will also be isotropic. As such, we can use the following relationship.

$$\phi = 4\pi\psi \quad (3)$$

3. Substitute the previous relationship in the scattering term.

$$\int_{4\pi} d\hat{\Omega}' \sigma_s(\mathbf{r}, \hat{\Omega}' \cdot \hat{\Omega}) \psi_p(\mathbf{r}, \hat{\Omega}') = \int_{4\pi} d\hat{\Omega}' \sigma_s(\mathbf{r}, \hat{\Omega}' \cdot \hat{\Omega}) \frac{\phi_p(\mathbf{r})}{4\pi} = \sigma_s(\mathbf{r}) \frac{\phi_p(\mathbf{r})}{4\pi} \quad (4)$$

4. Substituting the new scattering term back into the previous equation.

$$\hat{\Omega} \cdot \nabla \psi(\mathbf{r}, \hat{\Omega}) + \sigma(\mathbf{r})\psi(\mathbf{r}, \hat{\Omega}) = \sigma_s(\mathbf{r}) \frac{\phi_p(\mathbf{r})}{4\pi} + \frac{S_0}{4\pi} \delta(\mathbf{r} - \mathbf{r}_0) \quad (5)$$

5. Next, let's do a reflection of the previous equation such that $\hat{\Omega} \Rightarrow -\hat{\Omega}$

$$-\hat{\Omega} \cdot \nabla \psi(\mathbf{r}, \hat{\Omega}) + \sigma(\mathbf{r})\psi(\mathbf{r}, -\hat{\Omega}) = \sigma_s(\mathbf{r}) \frac{\phi_p(\mathbf{r})}{4\pi} + \frac{S_0}{4\pi} \delta(\mathbf{r} - \mathbf{r}_0) \quad (6)$$

6. Because the flux is isotropic and we have a point source, we know the following

$$\psi^\dagger(\mathbf{r}, \hat{\Omega}) = \psi(\mathbf{r}, -\hat{\Omega}) \quad (7)$$

7. Which implies

$$\int_{4\pi} \hat{\Omega} \psi(\mathbf{r}, -\hat{\Omega}) = \int_{4\pi} \hat{\Omega} \psi^\dagger(\mathbf{r}, \hat{\Omega}) \quad (8)$$

$$\frac{\phi_p(\mathbf{r})}{4\pi} = \frac{\phi_p^\dagger(\mathbf{r})}{4\pi} \quad (9)$$

$$\phi_p(\mathbf{r}) = \phi_p^\dagger(\mathbf{r}) \quad (10)$$

8. Substitute the previous relation into the equation we have been manipulating

$$-\hat{\Omega} \cdot \nabla \psi^\dagger(\mathbf{r}, \hat{\Omega}) + \sigma(\mathbf{r})\psi^\dagger(\mathbf{r}, \hat{\Omega}) = \sigma_s(\mathbf{r}) \frac{\phi_p(\mathbf{r})}{4\pi} + \frac{S_0}{4\pi} \delta(\mathbf{r} - \mathbf{r}_0) \quad (11)$$

9. When we move the scattering term to the left-hand side, we see the left-hand side becomes the adjoint equation

$$-\hat{\Omega} \cdot \nabla \psi^\dagger(\mathbf{r}, \hat{\Omega}) + \sigma(\mathbf{r})\psi^\dagger(\mathbf{r}, \hat{\Omega}) - \sigma_s(\mathbf{r}) \frac{\phi_p(\mathbf{r})}{4\pi} = H^\dagger \psi^\dagger(\mathbf{r}, \hat{\Omega}) = \frac{S_0}{4\pi} \delta(\mathbf{r} - \mathbf{r}_0) \quad (12)$$

10. From this, we see the adjoint equation is equal to the forward source, which, by extension, is also equal to the adjoint source.

$$H^\dagger \psi^\dagger(\mathbf{r}, \hat{\Omega}) = \frac{S_0}{4\pi} \delta(\mathbf{r} - \mathbf{r}_0) = q^\dagger \quad (13)$$

Inner Products

1. To begin, we will take the inner product over all independent variables of adjoint, point-source flux and the forward, distributed-source problem.

$$\langle \psi_p^\dagger H \psi_d \rangle = \langle \psi_p^\dagger q_d \rangle \quad (14)$$

$$\int_{4\pi} d\hat{\Omega} \int_V d\mathbf{V} \left[\psi_p^\dagger(\mathbf{r}, \hat{\Omega}) \hat{\Omega} \cdot \nabla \psi_d(\mathbf{r}, \hat{\Omega}) + \psi_p^\dagger(\mathbf{r}, \hat{\Omega}) \sigma(\mathbf{r}) \psi_d(\mathbf{r}, \hat{\Omega}) - \psi_p^\dagger(\mathbf{r}, \hat{\Omega}) \sigma_s(\mathbf{r}) \frac{\phi_d(\mathbf{r})}{4\pi} \right] = \int_{4\pi} d\hat{\Omega} \int_V d\mathbf{V} \psi_p^\dagger(\mathbf{r}, \hat{\Omega}) \frac{S_0}{V} \quad (15)$$

2. Next, we will take the inner product of the forward, distributed-source flux and the adjoint, point-source problem.

$$\langle \psi_d H^\dagger \psi_p^\dagger \rangle = \langle \psi_d q_d^\dagger \rangle \quad (16)$$

$$\int_{4\pi} d\hat{\Omega} \int_V d\mathbf{V} \left[-\psi_d(\mathbf{r}, \hat{\Omega}) \hat{\Omega} \cdot \nabla \psi_p^\dagger(\mathbf{r}, \hat{\Omega}) + \psi_d(\mathbf{r}, \hat{\Omega}) \sigma(\mathbf{r}) \psi_p^\dagger(\mathbf{r}, \hat{\Omega}) - \psi_d(\mathbf{r}, \hat{\Omega}) \sigma_s(\mathbf{r}) \frac{\phi_p^\dagger(\mathbf{r})}{4\pi} \right] = \int_{4\pi} d\hat{\Omega} \int_V d\mathbf{V} \psi_d(\mathbf{r}, \hat{\Omega}) q_p^\dagger \quad (17)$$

3. From Eq. 13, we know the adjoint source is equal to the forward source for the point-source problem.

$$\int_{4\pi} d\hat{\Omega} \int_V d\mathbf{V} \psi_d(\mathbf{r}, \hat{\Omega}) q_p^\dagger = \int_{4\pi} d\hat{\Omega} \int_V d\mathbf{V} \psi_d(\mathbf{r}, \hat{\Omega}) \frac{S_0}{4\pi} \delta(\mathbf{r} - \mathbf{r}_0) \quad (18)$$

4. Which, then becomes

$$\int_{4\pi} d\hat{\Omega} \int_V d\mathbf{V} \psi_d(\mathbf{r}, \hat{\Omega}) \frac{S_0}{4\pi} \delta(\mathbf{r} - \mathbf{r}_0) = \int_{4\pi} d\hat{\Omega} \psi_d(\mathbf{r}_0, \hat{\Omega}) \frac{S_0}{4\pi} = \phi_d(\mathbf{r}_0) \frac{S_0}{4\pi} \quad (19)$$

5. Plug the previous result into the previous equality relating inner products.

$$\int_{4\pi} d\hat{\Omega} \int_V d\mathbf{V} \left[-\psi_d(\mathbf{r}, \hat{\Omega}) \hat{\Omega} \cdot \nabla \psi_p^\dagger(\mathbf{r}, \hat{\Omega}) + \psi_d(\mathbf{r}, \hat{\Omega}) \sigma(\mathbf{r}) \psi_p^\dagger(\mathbf{r}, \hat{\Omega}) - \psi_d(\mathbf{r}, \hat{\Omega}) \sigma_s(\mathbf{r}) \frac{\phi_p^\dagger(\mathbf{r})}{4\pi} \right] = \phi_d(\mathbf{r}_0) \frac{S_0}{4\pi} \quad (20)$$

Difference of Inner Products

1. Next, we will subtract Eq. 20 from Eq. 15.

$$\begin{aligned} & \int_{4\pi} d\hat{\Omega} \int_V d\mathbf{V} \left[\psi_p^\dagger(\mathbf{r}, \hat{\Omega}) \hat{\Omega} \cdot \nabla \psi_d(\mathbf{r}, \hat{\Omega}) \right] + \int_{4\pi} d\hat{\Omega} \int_V d\mathbf{V} \left[\psi_d(\mathbf{r}, \hat{\Omega}) \hat{\Omega} \cdot \nabla \psi_p^\dagger(\mathbf{r}, \hat{\Omega}) \right] \\ & + \int_{4\pi} d\hat{\Omega} \int_V d\mathbf{V} \left[\psi_p^\dagger(\mathbf{r}, \hat{\Omega}) \sigma(\mathbf{r}) \psi_d(\mathbf{r}, \hat{\Omega}) \right] - \int_{4\pi} d\hat{\Omega} \int_V d\mathbf{V} \left[\psi_d(\mathbf{r}, \hat{\Omega}) \sigma(\mathbf{r}) \psi_p^\dagger(\mathbf{r}, \hat{\Omega}) \right] \\ & - \int_{4\pi} d\hat{\Omega} \int_V d\mathbf{V} \left[\psi_p^\dagger(\mathbf{r}, \hat{\Omega}) \sigma_s(\mathbf{r}) \frac{\phi_d(\mathbf{r})}{4\pi} \right] + \int_{4\pi} d\hat{\Omega} \int_V d\mathbf{V} \left[\psi_d(\mathbf{r}, \hat{\Omega}) \sigma_s(\mathbf{r}) \frac{\phi_p^\dagger(\mathbf{r})}{4\pi} \right] \\ & = \int_{4\pi} d\hat{\Omega} \int_V d\mathbf{V} \psi_p^\dagger(\mathbf{r}, \hat{\Omega}) \frac{S_0}{V} - \phi_d(\mathbf{r}_0) \frac{S_0}{4\pi} \end{aligned} \quad (21)$$

2. We clearly see the collision and scattering terms cancel each other out.

$$\int_{4\pi} d\hat{\Omega} \int_V d\mathbf{V} \left[\psi_p^\dagger(\mathbf{r}, \hat{\Omega}) \sigma(\mathbf{r}) \psi_d(\mathbf{r}, \hat{\Omega}) \right] - \int_{4\pi} d\hat{\Omega} \int_V d\mathbf{V} \left[\psi_d(\mathbf{r}, \hat{\Omega}) \sigma(\mathbf{r}) \psi_p^\dagger(\mathbf{r}, \hat{\Omega}) \right] = 0 \quad (22)$$

$$- \int_{4\pi} d\hat{\Omega} \int_V d\mathbf{V} \left[\psi_p^\dagger(\mathbf{r}, \hat{\Omega}) \sigma_s(\mathbf{r}) \frac{\phi_d(\mathbf{r})}{4\pi} \right] + \int_{4\pi} d\hat{\Omega} \int_V d\mathbf{V} \left[\psi_d(\mathbf{r}, \hat{\Omega}) \sigma_s(\mathbf{r}) \frac{\phi_p^\dagger(\mathbf{r})}{4\pi} \right] = 0 \quad (23)$$

3. Which leaves the difference of the two equalities as:

$$\begin{aligned} \int_{4\pi} d\hat{\Omega} \int_V d\mathbf{V} \left[\psi_p^\dagger(\mathbf{r}, \hat{\Omega}) \hat{\Omega} \cdot \nabla \psi_d(\mathbf{r}, \hat{\Omega}) \right] + \int_{4\pi} d\hat{\Omega} \int_V d\mathbf{V} \left[\psi_d(\mathbf{r}, \hat{\Omega}) \hat{\Omega} \cdot \nabla \psi_p^\dagger(\mathbf{r}, \hat{\Omega}) \right] \\ = \int_{4\pi} d\hat{\Omega} \int_V d\mathbf{V} \psi_p^\dagger(\mathbf{r}, \hat{\Omega}) \frac{S_0}{V} - \phi_d(\mathbf{r}_0) \frac{S_0}{4\pi} \end{aligned} \quad (24)$$

4. To cancel the streaming terms, we need to recognize the transport operator (H) is a linear operator, which can be written as the sum of a streaming term (L), collision term (C), and scattering term (S).

(a) We can expand the transport operator as such:

$$H\psi = q \quad \Rightarrow \quad [L + C - S]\psi = L\psi + C\psi - S\psi = q \quad (25)$$

(b) We have the adjoint identity

$$\langle \zeta^\dagger H \zeta \rangle = \langle \zeta H^\dagger \zeta^\dagger \rangle \quad (26)$$

(c) As the transport operator is linear, we can distributed the operators that make up the transport operator.

$$\langle \zeta^\dagger L \zeta \rangle + \langle \zeta^\dagger C \zeta \rangle - \langle \zeta^\dagger S \zeta \rangle = \langle \zeta L^\dagger \zeta^\dagger \rangle + \langle \zeta C^\dagger \zeta^\dagger \rangle - \langle \zeta S^\dagger \zeta^\dagger \rangle \quad (27)$$

(d) However, we are concerned with the streaming relationship, which can be written as follows because of the linear independence of the streaming, collision, and scattering operators.

$$\langle \zeta^\dagger L \zeta \rangle = \langle \zeta L^\dagger \zeta^\dagger \rangle \quad (28)$$

(e) Or, more generally,

$$\langle \zeta^\dagger L \xi \rangle = \langle \xi L^\dagger \zeta^\dagger \rangle \quad (29)$$

5. Next, we can recognize the streaming terms from Eq. 24 as the following inner products,

$$\int_{4\pi} d\hat{\Omega} \int_V d\mathbf{V} \left[\psi_p^\dagger(\mathbf{r}, \hat{\Omega}) \hat{\Omega} \cdot \nabla \psi_d(\mathbf{r}, \hat{\Omega}) \right] = \langle \psi_p^\dagger L \psi_d \rangle \quad (30)$$

$$\int_{4\pi} d\hat{\Omega} \int_V d\mathbf{V} \left[\psi_d(\mathbf{r}, \hat{\Omega}) \hat{\Omega} \cdot \nabla \psi_p^\dagger(\mathbf{r}, \hat{\Omega}) \right] = -\langle \psi_d L^\dagger \psi_p^\dagger \rangle \quad (31)$$

6. Which lets us rewrite Eq. 24,

$$\langle \psi_p^\dagger L \psi_d \rangle - \langle \psi_d L^\dagger \psi_p^\dagger \rangle = \int_{4\pi} d\hat{\Omega} \int_V d\mathbf{V} \psi_p^\dagger(\mathbf{r}, \hat{\Omega}) \frac{S_0}{V} - \phi_d(\mathbf{r}_0) \frac{S_0}{4\pi} \quad (32)$$

7. Using the adjoint identity from Eq. 29, we recognize the streaming terms will cancel out

$$\langle \psi_p^\dagger L \psi_d \rangle = \langle \psi_d L^\dagger \psi_p^\dagger \rangle \quad (33)$$

$$\therefore \langle \psi_p^\dagger L \psi_d \rangle - \langle \psi_d L^\dagger \psi_p^\dagger \rangle = 0 \quad (34)$$

8. Thus, we are left with the following expression.

$$0 = \int_{4\pi} d\hat{\Omega} \int_V d\mathbf{V} \psi_p^\dagger(\mathbf{r}, \hat{\Omega}) \frac{S_0}{V} - \phi_d(\mathbf{r}_0) \frac{S_0}{4\pi} \quad (35)$$

9. Using the relationship from Eq. 7, we can rewrite the integral from the previous expression

$$\int_{4\pi} d\hat{\Omega} \int_V d\mathbf{V} \psi_p^\dagger(\mathbf{r}, \hat{\Omega}) \frac{S_0}{V} = \int_{4\pi} d\hat{\Omega} \int_V d\mathbf{V} \psi_p(\mathbf{r}, \hat{\Omega}) \frac{S_0}{V} = \frac{S_0}{4\pi V} \int_V d\mathbf{V} \phi_p(\mathbf{r}) \quad (36)$$

10. We know the integral of the scalar flux over the entire spatial domain divided by the volume is the average scalar flux over the entire spatial domain.

$$\frac{1}{V} \int_V d\mathbf{V} \phi_p(\mathbf{r}) = \bar{\phi}_p \quad (37)$$

11. Substitute the previous two expressions back into Eq. 35.

$$0 = \bar{\phi}_p \frac{S_0}{4\pi} - \phi_d(\mathbf{r}_0) \frac{S_0}{4\pi} \quad (38)$$

$$\bar{\phi}_p \frac{S_0}{4\pi} = \phi_d(\mathbf{r}_0) \frac{S_0}{4\pi} \quad (39)$$

12. Finally, we see the average scalar flux in the solid due to a point isotropic source at $\mathbf{r}_0$ emitting S_0 neutrons/sec ($\bar{\phi}_p$) is equal to the scalar flux at $\mathbf{r}_0$ due to a source of total strength S_0 neutrons/sec uniformly and isotopically distributed throughout the solid ($\phi_d(\mathbf{r}_0)$).

$$\bar{\phi}_p = \phi_d(\mathbf{r}_0) \tag{40}$$

2 Rate of Convergence of Inner Iterations

A Equal Scattering and Total XS

$\Sigma_t = \Sigma_s$	# Inner Iterations
1	9
10	81
100	882

The number of inner iterations increases as the optical thickness of the medium increases. The number of inner iterations increases because the neutrons born in the center of the cell travel more mean free paths before their information reaches the cell boundary, so more collisions will occur with each additional collision requiring the calculation of an additional scattering source (1st, 2nd, 3rd, ... 882nd).

B Absorbing Medium

Σ_s	# Inner Iterations
10	4
90	47
100	882

The number of inner iterations increases with increased scattering. The number of inner iterations increases because a higher scattering ratio means a higher percent of neutrons will be included in the next scattering source, which requires the calculation of additional scattering sources because the previous scattering source is relatively larger.

3 Accuracy of the Diamond Difference Method

A "Very Brief" Explanation

To obtain these point values, we first call ‘sweep’ directly and forgo any inner iterations as the medium is purely absorbing. We also modify ‘sweep’ such that we only solve for the flux where $\mu, \eta > 0$ by deleting the other three loops. Once we have the cell-centered flux values, we calculate the face-averaged flux values. We start by marching in from the vacuum boundary where the face touching the vacuum has a flux value equal to half the corresponding cell centered value. After we have all the face-averaged values, we find the cell the point under consideration is located in, then perform a 2D linear interpolation using the cell center and adding weighted contributions from the face fluxes based on the relative location to the cell center. In mathematical form, we will implement the following equation in our:

$$\psi_{DD}(x, y) = \psi_{ij} + \frac{x - x_i}{\Delta x} (\psi_{i-1/2, j} - \psi_{i+1/2, j}) + \frac{y - y_i}{\Delta y} (\psi_{i, j-1/2} - \psi_{i, j+1/2}) \quad (41)$$

where

1. $\psi_{DD}(x, y)$ is the flux at a given point calculated via the diamond difference method,
2. ψ_{ij} is the cell-centered flux of the cell the point is located in,
3. $\psi_{i\pm 1/2, j}, \psi_{i, j\pm 1/2}$ are face averaged flux values,
4. (x, y) is the point under consideration,
5. (x_i, y_i) is the location of the cell center,
6. $\Delta x, \Delta y$ are the cell sizes in the x and y directions respectively.

I should note we are not directly interpolating between the cell centered fluxes because the point (0.05, 0.95) is in the cell in the top left corner for the 10x10 case. Therefore, we have to calculate and use the face-averaged fluxes

For our case, the cell-averaged flux values are the same as the cell-centered flux values. We see this when looking at the formula for cell averaged flux:

$$\psi_{i, j} = \frac{\psi_{in, x} + \psi_{out, x}}{2} = \frac{\psi_{in, y} + \psi_{out, y}}{2} \quad (42)$$

The cell-averaged flux is the average value between the two faces giving both faces equal weight, which is the same as the cell-centered flux values.

B Approximate Values

Using the same number of sig figs as the exact flux.

Table 1: Approximate Flux Values at Specific Points

Point	10x10	30x30	50x50
(0.35, 0.65)	0.453403	0.454454	0.454541
(0.05, 0.95)	0.0797003	0.0825982	0.0828288
(0.15, 0.15)	0.197327	0.218892	0.222921

C Error in the Diamond Difference Solution

Table 2: Error in the Diamond Difference Solution

Point	10x10	30x30	50x50
(0.35, 0.65)	0.001187	0.001187	0.001187
(0.05, 0.95)	0.003259	0.0032592	0.003259
(0.15, 0.15)	0.031473	0.031473	0.031473

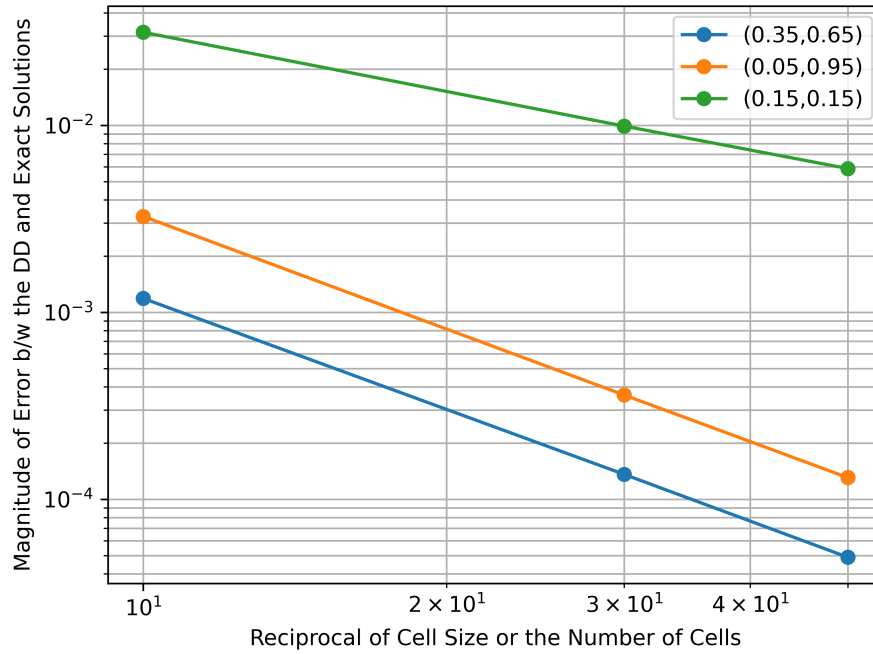


Figure 1: Error b/w the Diamond Difference and Exact Solutions vs 1/Cell Size

D Order of Error

The order of the error, p , for each point can be calculated using the following equation:

$$p \approx \frac{\log\left(\frac{err_{10 \times 10}}{err_{50 \times 50}}\right)}{\log\left(\frac{1/mesh \ size_{10 \times 10}}{1/mesh \ size_{50 \times 50}}\right)} \quad (43)$$

Because we are interested in asymptotic convergence, I am using the errors from the 30x30 and 50x50 meshes instead of finding the best line of fit between all three points.

Plugging in the errors we calculated in the previous part yields the following orders of convergence:

Table 3: Order of Error for each Point

Point	Convergence Order
(0.35, 0.65)	1.99840
(0.05, 0.95)	1.99051
(0.15, 0.15)	1.02179

E Comparison to Madsen's Result

Madsen's result is given by Theorem 2, which states error decreases quadratically with decreasing mesh size.

$$\|\epsilon\| \leq C \|\psi\|_3 h^2 \quad (44)$$

where

1. ϵ is the error,
2. C is some constant,
3. $\|\psi\|_3$ denotes the magnitude of the flux using a third order Taylor expansion,
4. h is the mesh size.

For the points (0.35, 0.65) and (0.05, 0.95), we see the convergence orders are both roughly 2, which agrees with Madsen's result. However, the point (0.15, 0.15) has a convergence order of 1, which disagrees with Madsen's result. Madsen's theorem 2 requires the second derivative to be continuous along the domain. However, our flux only has a continuous first derivative, aside from along characteristics where the first derivative is not continuous, so our problem does not meet the requirements for the theorem. The convergence order of (0.15, 0.15) is only 1 because it lies on a characteristic, which means the derivative in flux will

be discontinuous. When not on a characteristic, the error in the diamond difference approximation is proportional to the discontinuous second derivative of the flux (the error bounding term in the first-order Taylor expansion of flux). However, when, the first derivative is discontinuous, the error is bounded by the discontinuous first derivative resulting in first order convergence.